



Department of Computer Science and Engineering

Academic Year 2024– 2025 (Odd Semester)

Degree, Semester & Branch: BE,III Semester B.E. ECE

Course Code & Title: CS3353 & C Programming and Data Structures

Name of the Faculty member (s): K.Priya, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit I / Functions
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO2
- **Activity Chosen:** Crossword Puzzle
- **Justification:**
 - A crossword puzzle is a word puzzle that usually consists of a white and black-shaded squares. By solving clues that lead to the answers, the goal is to fill the white squares with letters, forming words or phrases.
 - A function is a set of statements that when called to perform some specific task. It is used to perform certain actions, and they are important for reusing code.
 - So, a crossword puzzle activity is chosen to have a better understanding of the Functions by finding the suitable word according to the question
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - Crossword puzzle of 11 questions were given to each students.
 - The students were understanding the questions, patterns and completed the puzzle as shown in Fig 1
 - After completion of the puzzle activity, the students were discussed among them and finally the faculty member discussed the answer to make the students aware of the correct answer as shown in Fig 2
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO1
CO1	3	2	1	1	2	1	2

(1 – Low 2 – Moderate 3 – High)



- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PO11	PSO1
Justification for correlation	Students will be able to use basic knowledge of engineering fundamentals to develop C programs	Students will be able to use basic C programming concepts in analysis of complex engineering problems	Students will be able to use basic C programming concepts in designing solutions for complex engineering problems	Student will be able to write C program individually using basic C Concepts.	Students will be able to execute and present programs using basic C Concepts	Student will become aware of the need for lifelong learning and the continued upgrading of technical knowledge using C Concepts	Student will be able to apply C programming concepts in emerging trends in Information and Communication Technology

- Images / Screenshot of the practice:

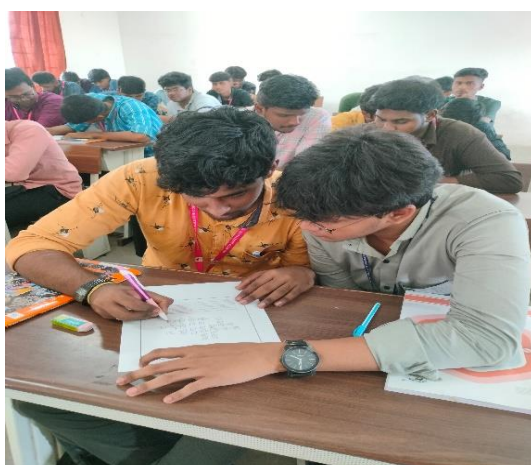


Fig 1: Students Solving Crossword Puzzle

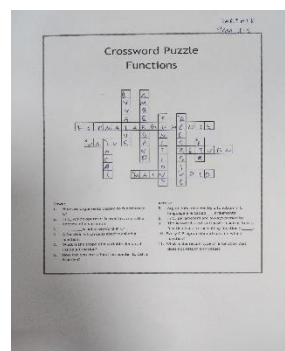
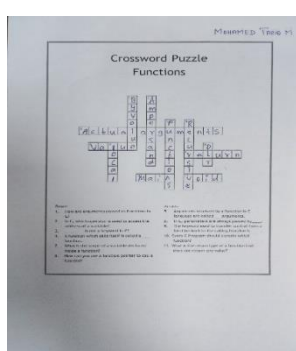
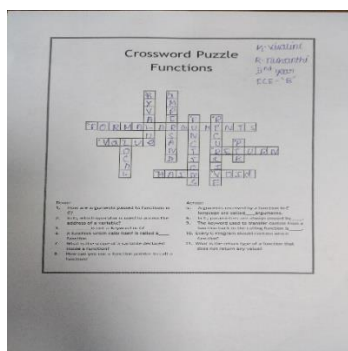


Fig 2: Sample Sheet of Solved Crossword Puzzle



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students said that the crossword puzzle activity was very interesting and they were able to identify the appropriate concepts related to Functions.
- They said the it will be helpful while preparing for examination

- ❖ ***Benefit of the practice:***

- Student's understanding of concepts Functions and how to reuse functions across different parts of the program to reduce the code duplication.
- Vocabulary of the terms related to Functions in C Programming was improved
- This activity helps to test the level of understanding of the students

- ❖ ***Challenges faced in implementation:***

- Few students found it difficult to complete the puzzle
- Planned for 15 minutes, but unable to be completed within 15 minutes, it will take an extra 5 minutes to complete.
- Students just might not have the necessary knowledge to complete crossword puzzles

References:

- <https://www.ritrjpm.ac.in/images/computer-science/Crosswords-MCAP-GM.pdf>
- <http://www.classtools.net/crossword/>
- <https://www.activityvillage.co.uk/crosswords>

Signature of Faculty Member

HOD-CSE